Cohen 2021 Multiple Myeloma — analysis summary

RUN 1 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

54 samples intermix per-neighbourhood (iLISI 1.0→14.1 of 54) on one continuous connected component; single Leiden lobe, aspect 1.09, only 2.8% cycling — no islands, no batch archipelago, no cycling compartment.

Gate 2 — module geography → GO (non-branching)

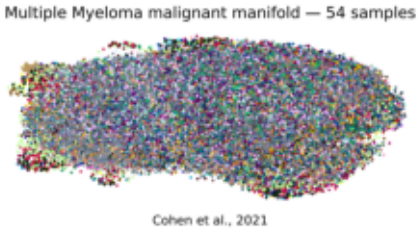
Non-branching: a single Gast/pSom-origin → terminal-Crest axis. Crest inversely opposes Gast (r=−0.69) and pSom (r=−0.61); pSom&Crest joint-high enrichment 0.22 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — hub cross-repression → NA

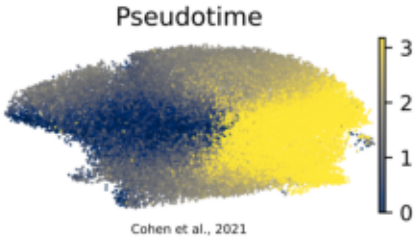
No bifurcation: Gast↔pSom near-orthogonal (r=+0.07), no competing arms — branching-only gate not applicable.

STEP FIGURES

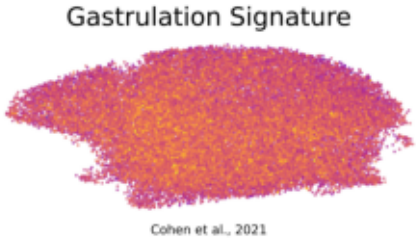
Manifold — samples



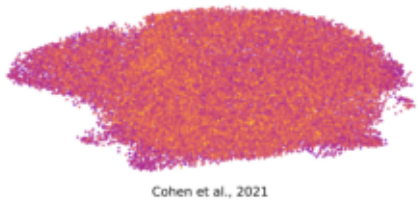
Pseudotime



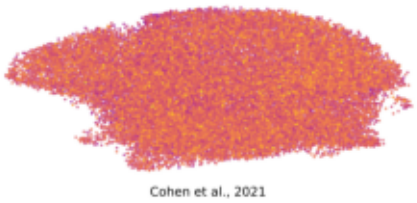
Gastrulation



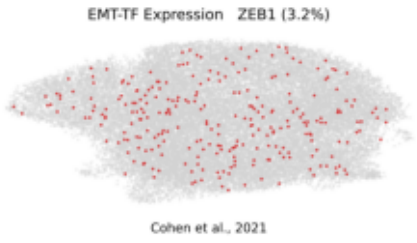
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Multiple myeloma bone-marrow plasma cells (Cohen et al., 2021), MARS-seq — 33,076 malignant cells across 54 samples (≥100 cells). Whole malignant manifold; single differentiation continuum.

Verdict.
The embryonic pattern reproduces as a NON-BRANCHING axis: Gast/pSom-high origin → terminal neural-crest, no resolved bifurcation. Linear-pSom/Crest geometry (Gast μ=−0.022, pSom μ=+0.014, Crest μ=+0.001). EMT-TF ZEB1 is the only factor above the detection floor.

Parameter choices

Sample threshold	≥100 cells (floor; smooth decline, no cliff) → 54/73 samples, 33,147 cells	**Overdispersed genes**	482 (scFates); island removal 71 cells / 2 specks → 33,076
Integration	Harmony θ=10 on sample (swept 2/5/10; same-sample nbr frac 0.91→0.11)	**Embedding**	n_pcs=20 (scree plateau), n_neighbors=30, UMAP spread=1.3
Trajectory	scFates curve (ElPiGraph, N=20, λ=0.3, μ=100); non-branching, fold-free (2.5)	**Root**	tip highest Gast+pSom, lowest Crest; Gast ρ=−0.55, Crest ρ=+0.57
EMT-TF readout	ZEB1 (3.2%) — sole factor above floor; others <0.3%	**Annotation**	Malignant plasma cells (author cell_type); single continuum